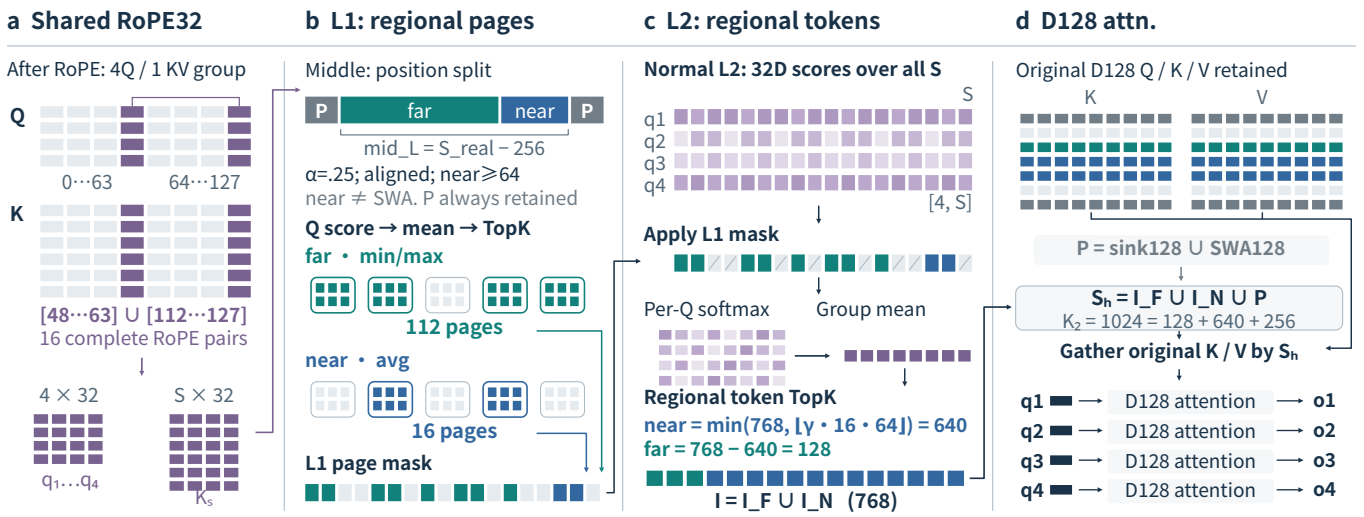

a Shared RoPE32
After RoPE: 4Q / 1 KV group
Q
0···63
64···127
K
[48···63] ∪ [112···127]
16 complete RoPE pairs
4 × 32
S × 32
q1…q4
Ks
b L1: regional pages
Middle: position split
P
far
near
P
mid_L = S_real − 256
α=.25; aligned; near⩾64
near ≠ SWA. P always retained
Q score → mean → TopK
far • min/max
112 pages
near • avg
16 pages
L1 page mask
c L2: regional tokens
Normal L2: 32D scores over all S
S
q1
q2
q3
q4
[4, S]
Apply L1 mask
Per-Q softmax
Group mean
Regional token TopK
near = min(768, ⌊γ • 16 • 64⌋) = 640
far = 768 − 640 = 128
I = I_F ∪ I_N (768)
d D128 attn.
Original D128 Q / K / V retained
K
V
P = sink128 ∪ SWA128
Sh = I_F ∪ I_N ∪ P
K2 = 1024 = 128 + 640 + 256
Gather original K / V by Sh
q1 → D128 attention → o1
q2 → D128 attention → o2
q3 → D128 attention → o3
q4 → D128 attention → o4
Same RoPE32 in L1 / L2
K1=128. β=.125 → near 16 / far 112 pages
4Q: same IDs, own attn.
Final attention weights